

All My Work

Your saved learning record for Molecular Genetics.

7 completed activity runs. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 0 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Recombinant DNA: different jobs in the pathway

9/17/2026, 8:41:41 PM • Elapsed 0m 0s • Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Restriction enzyme — Correct

Reference: Restriction enzyme

Label B

Attempt 1: Compatible ends — Correct

Reference: Compatible ends

Label C

Attempt 1: DNA ligase — Correct

Reference: DNA ligase

Label D

Attempt 1: Vector — Correct

Reference: Vector

Label E

Attempt 1: Host expression — Correct

Reference: Host expression

Label F

Attempt 1: Function testing — Correct

Reference: Function testing

DNA gel: read direction, size and patterns

9/17/2026, 8:41:41 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Loading wells — Correct

Reference: Loading wells

Label B

Attempt 1: Larger fragments — Correct

Reference: Larger fragments

Label C

Attempt 1: Smaller fragments — Correct

Reference: Smaller fragments

Label D

Attempt 1: DNA migration toward positive electrode — Correct

Reference: DNA migration toward positive electrode

Read the message in codons

9/17/2026, 8:41:41 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: mRNA codon — Correct

Reference: mRNA codon

Label B

Attempt 1: tRNA anticodon — Correct

Reference: tRNA anticodon

Label C

Attempt 1: Amino acid — Correct

Reference: Amino acid

Label D

Attempt 1: Stop codon — Correct

Reference: Stop codon

Label E

Attempt 1: Ribosome — Correct

Reference: Ribosome

Coding DNA, template DNA and mRNA

9/17/2026, 8:41:41 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Coding DNA strand — Correct

Reference: Coding DNA strand

Label B

Attempt 1: Template DNA strand — Correct

Reference: Template DNA strand

Label C

Attempt 1: Transcription — Correct

Reference: Transcription

Label D

Attempt 1: mRNA — Correct

Reference: mRNA

Replication products: semiconservative

9/17/2026, 8:41:40 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Parental strand — Correct

Reference: Parental strand

Label B

Attempt 1: New strand — Correct

Reference: New strand

Label C

Attempt 1: Parental strand — Correct

Reference: Parental strand

Label D

Attempt 1: New strand — Correct

Complementarity and antiparallel direction

9/17/2026, 8:41:40 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Sugar–phosphate backbone — Correct

Reference: Sugar–phosphate backbone

Label B

Attempt 1: Complementary base pair — Correct

Reference: Complementary base pair

Label C

Attempt 1: 3' end — Correct

Reference: 3' end

Label D

Attempt 1: 3' end — Correct

Reference: 3' end

Evidence that DNA carries genetic information

9/17/2026, 8:41:38 PM · Elapsed 0m 0s · Required check

Review submitted responses and attempts

First-try multiple choice: 1/2. Writing is not automatically graded.

Why did Hershey and Chase use different radioactive labels?

Options: To distinguish the movement of DNA from protein · To prevent bacteria from containing DNA · To measure chromosome number during meiosis · To make all viral molecules chemically identical

Attempt 1: To prevent bacteria from containing DNA — Incorrect

Attempt 2: To distinguish the movement of DNA from protein — Correct

Which statement goes beyond Griffith's evidence alone?

Options: Transformation required further explanation · The transforming substance has a double-helical DNA structure · Some material from killed bacteria changed a living lineage · The acquired trait could persist in descendants

Attempt 1: The transforming substance has a double-helical DNA structure — Correct

Explain why destroying DNA in a transforming preparation provides stronger evidence than merely detecting DNA in the preparation.

Real-origin saved Chapter 18 work. Presence shows association, not necessity. If destroying DNA removes transformation while appropriate comparison treatments retain it, the result links intact DNA with the measured transforming function. The conclusion remains tied to the assay and its controls.

Submitted writing; human review required.

Compare the questions addressed by Griffith and Hershey–Chase.

Griffith asked whether a heritable change could be transferred between bacterial preparations. Hershey and Chase traced DNA and protein during bacteriophage infection to determine which material was more closely associated with infected cells. Their experiments address different stages in identifying hereditary material.

Submitted writing; human review required.

Frayer work

No Frayer words selected.

Collected optional responses

No optional written responses collected.

Textbook question work

Draft and saved question responses. Optional and ungraded.

In Text · p. 625 · question 1 — Saved

Real-origin saved Chapter 18 work. Presence shows association, not necessity. If destroying DNA removes transformation while appropriate comparison treatments retain it, the result links intact DNA with the measured transforming function. The conclusion remains tied to the assay and its controls. Edited.

Supported and independent practice